Complex Analysis Homework 4

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15. Use the Cauchy inequalities or the maximum modulus principle to solve the following problems:
(a) Prove that if f is an entire function that satisfies

$$_{|z|=R}|f(z)| \leq AR^k + B$$

for all $R > 0$, and for some integer $k \geq 0$ and some constants $A, B > 0$, then f is a polynomial of degree $\leq k$.
(b) Show that if f is holomorphic in the unit disc, is bounded, and converges uniformly to zero in the sector $\theta < \arg z < \varphi$ as $|z| \to 1$, then $f = 0$.
(c) Let $w_1, \ldots, w_n$ be points on the unit circle in the complex plane. Prove that there exists a point z on the unit circle such that the product of the distances from z to the points w_j, $1 \leq j \leq n$, is at least 1. Conclude that there exists a point w on the unit circle such that the product of the distances from w to the points w_j, $1 \leq j \leq n$, is exactly equal to 1.
(d) Show that if the real part of an entire function f is bounded, then f is constant.

证明 *(a)* 由柯西不等式,$x < R$ 时

$$|f^{(n)}(x)| \leq \frac{n!}{R^n} \sup_{|\zeta|=R} |f(\zeta)| \leq n! \frac{AR^k + B}{R^n}.$$

当 $n > k$ 时，令 $R \to \infty$，右边趋于 0，故 $f^{(n)} = 0$。因此 f 只有次数 $\leq k$ 的项，即为次数 $\leq k$ 的多项式。
(b) 考虑辅助函数

$$g(z) = f(z) f\left(ze^{i\frac{\varphi-\theta}{2}}\right) f\left(ze^{i\frac{2(\varphi-\theta)}{2}}\right) \cdots f\left(ze^{i\frac{m(\varphi-\theta)}{2}}\right),$$

其中 m 足够大使得 $\frac{m(\varphi-\theta)}{2}>2\pi$，故而 g 在整个单位圆周 S^1 上均一致趋于 *0*，由最大模原理有 $g\equiv 0$，故而 f 在单位圆盘上的零点不可数，进而 $f\equiv 0$.

(c) 考虑多项式

$$P(z)=\prod_{j=1}^{n}(z-w_j).$$

在单位圆周上 $|z|=1$，故

$$|P(z)|=\prod_{j=1}^{n}|z-w_j|.$$

由最大模原理，$|P(z)|$ 在闭单位圆盘上的最大值在边界取得。而

$$|P(0)|=\prod_{j=1}^{n}|w_j|=1.$$

因此边界上必有一点 z 使得 $|P(z)|\geq|P(0)|=1$。

函数 $\prod|z-w_j|$ 在单位圆周上连续，而 $P(w_1)=0$，由介值定理，必存在点使乘积恰好等于 *1*。

(d) 设 $\operatorname{Re} f(z)\leq M$。令

$$g(z)=e^{f(z)}.$$

则

$$|g(z)|=e^{\operatorname{Re} f(z)}\leq e^{M}.$$

g 是有界整函数，由刘维尔定理，g 为常数，从而 f 为常数。

16. Suppose f and g are holomorphic in a region containing the disc $|z|\leq 1$. Suppose that f has a simple zero at $z=0$ and vanishes nowhere else in $|z|\leq 1$. Let

$$f_\epsilon(z)=f(z)+\epsilon g(z).$$

Show that if ϵ is sufficiently small, then

(a) $f_\epsilon(z)$ has a unique zero in $|z|\leq 1$, and

(b) if z_ϵ is this zero, the mapping $\epsilon\mapsto z_\epsilon$ is continuous.

证明 *(a)* 由于 f 只有 $z=0$ 一个零点，故而在 S^1 上非零连续，设在 S^1 上有 $|f|\geq m>0$，再设在 S^1 上有 $|g|\leq M$，故而取 $\forall\epsilon<\frac{m}{M}$，则在 S^1 上有

$$|f|\geq m>\epsilon M\geq|\epsilon g|,$$

因此由 *Rouché* 定理，f 与 $f+\epsilon g$ 在单位圆盘内的零点个数相同.

(b) 记为 $f_\epsilon(z_\epsilon)=0$，设 $\epsilon_n\to\epsilon$，我们只需证明 $z_{\epsilon_n}\to z_\epsilon$.

首先由定义，我们有

$$\epsilon=\frac{f(z_\epsilon)}{g(z_\epsilon)},\quad \epsilon_n=\frac{f(z_{\epsilon_n})}{g(z_{\epsilon_n})},$$

而由 $\epsilon_n\to\epsilon$，有

$$\lim_{n\to\infty}\frac{f(z_{\epsilon_n})}{g(z_{\epsilon_n})}=\lim_{n\to\infty}\epsilon_n=\epsilon=\frac{f(z_\epsilon)}{g(z_\epsilon)}.$$

注意到 z_ϵ 是 f/g 的一阶 ϵ-点，即 z_ϵ 是 $f/g-\epsilon$ 的一阶零点，这是因为 $f+\epsilon g$ 的零点是唯一的. 故而在 z_ϵ 足够小的邻域上，f/g 是单射. 因此由

$$\frac{f(z_{\epsilon_n})}{g(z_{\epsilon_n})}\to\frac{f(z_\epsilon)}{g(z_\epsilon)},$$

结合 f/g 的连续性与单射性，有 $z_{\epsilon_n}\to z_\epsilon$.

17. Let f be non-constant and holomorphic in an open set containing the closed unit disc.

(a) Show that if $|f(z)|=1$ whenever $|z|=1$, then the image of f contains the unit disc. [Hint: One must show that $f(z)=w_0$ has a root for every $w_0\in\mathbb{D}$. To do this, it suffices to show that $f(z)=0$ has a root (why?). Use the maximum modulus principle to conclude.]

(b) If $|f(z)|\geq 1$ whenever $|z|=1$ and there exists a point $z_0\in\mathbb{D}$ such that $|f(z_0)|<1$, then the image of f contains the unit disc.

证明 *(a)* 根据 *Rouché* 定理，对于任意的 $w_0\in\mathbb{D}$，$f(z)-w_0$ 和 $f(z)$ 有相同的零点数，利用第二章练习 *2* 的 *15* 题，$f(z)$ 存在零点，进而 $f(z)-w_0$ 存在零点，得证

(b) 类似 *(a)* 只需要证明 $f(z)$ 存在零点，不妨设零点不存在，考虑函数 $g(z)=\dfrac{1}{f(z)}$，利用最大模原理，在单位圆盘内 $g(z)\leq 1$，从而 $f(z)\geq 1$，矛盾！所以零点存在

19. Prove the maximum principle for harmonic functions, that is:

(a) If u is a non-constant real-valued harmonic function in a region Ω, then u cannot attain a maximum (or a minimum) in Ω.

(b) Suppose that Ω is a region with compact closure $\overline{\Omega}$. If u is harmonic in Ω and continuous in $\overline{\Omega}$, then

$$_{z\in\Omega}|u(z)| \leq_{z\in\overline{\Omega}-\Omega} |u(z)|.$$

[Hint: To prove the first part, assume that u attains a local maximum at z_0. Let f be holomorphic near z_0 with $u = \Re(f)$, and show that f is not open. The second part follows directly from the first.]

证明 *(a)* 反证，不妨设 u 在 z_0 处取得极大值，根据 u 的调和性质于是可以取 f 在 z_0 附近时全纯的而且 $u = Re(f)$, 但由于 z_0 处 f 是实部极大，所以 f 不是开的，进而矛盾。得证

(b) 紧集上连续函数一定有最值，从而根据前一问自然有

$$sup_{z\in\Omega}|u(z)| \leq sup_{z\in\overline{\Omega}-\Omega}|u(z)|.$$

21. Certain sets have geometric properties that guarantee they are simply connected.

(a) An open set $\Omega \subset \mathbb{C}$ is convex if for any two points in Ω, the straight line segment between them is contained in Ω. Prove that a convex open set is simply connected.

(b) More generally, an open set $\Omega \subset \mathbb{C}$ is star-shaped if there exists a point $z_0 \in \Omega$ such that for any $z \in \Omega$, the straight line segment between z and z_0 is contained in Ω. Prove that a star-shaped open set is simply connected. Conclude that the slit plane $\mathbb{C} - \{(-\infty, 0]\}$ (and more generally any sector, convex or not) is simply connected.

(c) What are other examples of open sets that are simply connected?

证明 *(a)* 若凸集 Ω 不是单连通，则一定存在闭曲线 γ 不同伦于一个点，因此 γ 内部有个点不属于 Ω，而 γ 是环绕着这个点的，因此能在 γ 上取两个点，与该点三点共线，而 $\gamma \subset \Omega$，因此与凸集矛盾.

(b) 对于 Ω 内的任意闭合曲线 $p(s)$，我们作

$$H(r, s) = rz_0 + (1 - r)p(s), \quad r \in [0, 1].$$

由于 $p(s) \subset \Omega$ 并且 Ω 是关于 z_0 的星状区域，故而 $H(r, s) \in \Omega, \forall r, s$，并且易知 H 连续，而 $H(0, s) = p(s), H(1, s) = z_0$，即 $p(s)$ 同伦于 z_0 点，注意到 $p(s)$ 是任意的闭曲线，故而所有闭曲线均同伦于点，因此 Ω 单连通.

(c) 整个复平面

22. Show that there is no holomorphic function f in the unit disc $\mathbb{D}$ that extends continuously to $\partial\mathbb{D}$ such that $f(z) = 1/z$ for $z \in \partial\mathbb{D}$.

证明 不妨设 f 存在，考虑 $g(z) = zf(z) - 1$，在边界上恒为 0，从而根据最大模原理得到 $g(z) = 0$ $z \in \partial\mathbb{D}$，代入 $z = 0$，有 $1 = 0$，矛盾！

1. Consider a holomorphic map on the unit disc $f : \mathbb{D} \to \mathbb{C}$ which satisfies $f(0) = 0$. By the open mapping theorem, the image $f(\mathbb{D})$ contains a small disc centered at the origin. We then ask: does there exist $r > 0$ such that for all $f : \mathbb{D} \to \mathbb{C}$ with $f(0) = 0$, we have $D_r(0) \subset f(\mathbb{D})$?

(a) Show that with no further restrictions on f, no such r exists. It suffices to find a sequence of functions $\{f_n\}$ holomorphic in $\mathbb{D}$ such that $1/n \notin f(\mathbb{D})$. Compute $f_n'(0)$, and discuss.

(b) Assume in addition that f also satisfies $f'(0) = 1$. Show that despite this new assumption, there exists no $r > 0$ satisfying the desired condition. [Hint: Try $f_\epsilon(z) = \epsilon(e^{z/\epsilon} - 1)$.]

The Koebe-Bieberbach theorem states that if in addition to $f(0) = 0$ and $f'(0) = 1$ we also assume that f is injective, then such an r exists and the best possible value is $r = 1/4$.

(c) As a first step, show that if $h(z) = \frac{1}{z} + c_0 + c_1 z + c_2 z^2 + \cdots$ is analytic and injective for $0 < |z| < 1$, then $\sum_{n=1}^{\infty} n|c_n|^2 \leq 1$. [Hint: Calculate the area of the complement of $h(D_\rho(0) - \{0\})$ where $0 < \rho < 1$, and let $\rho \to 1$.]

(d) If $f(z) = z + a_2 z^2 + \cdots$ satisfies the hypotheses of the theorem, show that there exists another function g satisfying the hypotheses of the theorem such that $g^2(z) = f(z^2)$. [Hint: $f(z)/z$ is nowhere vanishing so there exists ψ such that $\psi^2(z) = f(z)/z$ and $\psi(0) = 1$. Check that $g(z) = z\psi(z^2)$ is injective.]

(e) With the notation of the previous part, show that $|a_2| \leq 2$, and that equality holds if and only if

$$f(z) = \frac{z}{(1 - e^{i\theta} z)^2} \quad \text{for some } \theta \in \mathbb{C}.$$

[Hint: What is the power series expansion of $1/g(z)$? Use part (c).]

(f) If $h(z) = \frac{1}{z} + c_0 + c_1 z + c_2 z^2 + \cdots$ is injective on $\mathbb{D}$ and avoids the values z_1 and z_2, show that $|z_1 - z_2| \leq 4$. [Hint: Look at the second coefficient in the power series expansion of $1/(h(z) - z_j)$.]

(g) Complete the proof of the theorem. [Hint: If f avoids w, then $1/f$ avoids 0 and $1/w$.]

证明 *(a)* 取 $f_n = \dfrac{z}{n}$, 则 $f_n'(0) = \dfrac{1}{n}$

(b) 取 $f_\epsilon(z) = \epsilon(e^{z/\epsilon} - 1)$, 则对于 $-\epsilon$, 无法取到

(c) 首先注意到 c_0 与本题条件和结论均无关, 因此不妨设 $c_0 = 0$. 再注意到, 如果用 $\lambda h(\lambda z)$ 代替 $h(z)$, 其中 $|\lambda| = 1$, 本题条件结论不变, 故而不妨再设 $c_1 \in \mathbb{R}$.

我们先设

$$U_r = \{z : |z| < r\}, \quad C_r = \{z : |z| = r\}, \quad V_r = \{z : r < |z| < 1\},$$

其中 $r \in (0, 1)$. 由开映射定理, $h(U_r)$ 是 ∞ 的某个邻域, 即 $\mathbb{C} \setminus h(U_r)$ 是 $\mathbb{C}$ 上的紧集, 再由 h 是单射知 $h(U_r), h(C_r), h(V_r)$ 互不相交. 我们令

$$h(z) = \frac{1}{z} + c_1 z + \varphi(z), \quad z \in \mathbb{D},$$

再设 $h = u + iv$, 作 $z = re^{i\theta}$, 则有

$$u = A\cos\theta + \operatorname{Re}\varphi, \quad v = -B\sin\theta + \operatorname{Im}\varphi,$$

其中

$$A = \frac{1}{r} + c_1 r, \quad B = \frac{1}{r} - c_1 r.$$

于是有

$$\frac{u^2}{A^2} + \frac{v^2}{B^2} = 1 + \frac{2\cos\theta}{A}\operatorname{Re}\varphi + \left(\frac{\operatorname{Re}\varphi}{A}\right)^2 - \frac{2\sin\theta}{B}\operatorname{Im}\varphi + \left(\frac{\operatorname{Im}\varphi}{B}\right)^2.$$

注意到 φ 在 $z = 0$ 处至少是二阶零点, 即

$$\varphi = O(r^2), \quad r \to 0,$$

而由 A, B 表达式知

$$\frac{1}{A} = O(r), \quad \frac{1}{B} = O(r), \quad r \to 0,$$

故而有

$$\frac{u^2}{A^2} + \frac{v^2}{B^2} = 1 + O(r^3), \quad r \to 0.$$

因此, $\exists \eta > 0$ 使得对于任意充分小的 r 均有

$$\frac{u^2}{A^2} + \frac{v^2}{B^2} < 1 + \eta r^3,$$

其中 $z = re^{i\theta}$. 也就是说, $h(C_r)$ 的像包含在椭圆

$$E_r : \frac{u^2}{A^2} + \frac{v^2}{B^2} = 1 + \eta r^3$$

内部, 进而有 $E_r \subset h(U_r)$, $h(V_r)$ 也在 E_r 内部.

一方面, 椭圆 E_r 所围面积为

$$\pi AB(1 + \eta r^3) = \pi \left(\frac{1}{r} + c_1 r\right)\left(\frac{1}{r} - c_1 r\right)(1 + \eta r^3) \le \frac{\pi}{r^2}(1 + \eta r^3).$$

另一方面, $h(V_r)$ 的面积为

$$\begin{aligned}
|h(V_r)| &= \iint_{V_r} |h'(z)|^2 dxdy \\
&= \int_r^1 dt \int_0^{2\pi} \left(-\frac{1}{t^2}e^{-2i\theta} + \sum_{k=1}^{\infty} c_k t^{k-1} e^{i(k-1)\theta} k\right)\left(-\frac{1}{t^2}e^{2i\theta} + \sum_{k=1}^{\infty} \overline{c_k} t^{k-1} e^{-i(k-1)\theta} k\right) t d\theta \\
&= 2\pi \int_r^1 \left(\frac{1}{t^3} + \sum_{k=1}^{\infty} k^2 |c_k|^2 t^{2k-1}\right) dt \\
&= \pi \left(\frac{1}{r^2} - 1 + \sum_{k=1}^{\infty} k|c_k|^2 (1 - r^{2k})\right).
\end{aligned}$$

于是乎, 我们有

$$\frac{\pi}{r^2}(1 + \eta r^3) \ge \pi \left(\frac{1}{r^2} - 1 + \sum_{k=1}^{\infty} k|c_k|^2 (1 - r^{2k})\right),$$

化简有

$$\sum_{n=1}^{\infty} n|c_n|^2 (1 - r^{2n}) \le 1 + \eta r,$$

由 $0 < r < 1$ 的任意性, 令 $r \to 0$ 即得

(d) 由于 $f(z)/z$ 没有零点, 所以存在 ψ 使得 $\psi^2(z) = f(z)/z$, 设

$$g(z) = z\psi(z^2)$$

计算得 $g(0) = 0, g'(0) = 1$

如果 $g(z)=g(w)$，则自然有 $f(z^2)=f(w^2)$，若 $z\neq w$，则 $z=-w$，从而

$$g(z)=z\psi(z^2)=-w\psi^2(w^2)=-g(w)$$

，进而 $g(z)=g(w)=0$，从而 $f(z)=f(w)=0$，于是 $z=w$，综上 g 是单射

(e) 计算得

$$g(z)=z(1+\frac{1}{2}a_2z^2+\cdots)$$

设 $h(z)=\dfrac{1}{g(z)}$，于是

$$h(z)=\frac{1}{z}(1-\frac{1}{2}a_2z^2+\cdots)$$

根据 g 的性质可得 h 满足 c 的条件，利用 c 有

$$\frac{1}{2}|a_2|\leq 1$$

即 $|a_2|\leq 2$，等号成立仅当 $h(z)=\dfrac{1}{z}(1-\dfrac{1}{2}a_2z^2)$ 且 $a_2=2e^{i\theta}$ 于是 $g(z)=\dfrac{z}{1-e^{i\theta}z^2}$，从而

$$f(z)=\frac{z}{(1-e^{i\theta}z)^2}\quad \textit{for some } \theta\in\mathbb{C}.$$

(f) 考虑函数 $\dfrac{1}{h(z)-z_j}$，其中 $j=1,2$. 由条件 $\dfrac{1}{h(z)-z_j}$ 全纯，并且有

$$\frac{1}{h(z)-z_j}=z-(c_0-z_j)z^2+\cdots.$$

于是

$$\frac{1}{h(0)-z_j}=0,\quad \left(\frac{1}{h(z)-z_j}\right)'\bigg|_{z=0}=1.$$

由 *(e)* 的结论，有

$$|c_0-z_j|\leq 2,\quad j=1,2,$$

故而有

$$|z_1-z_2|\leq|c_0-z_1|+|c_0-z_2|\leq 4.$$

(g) 设单射全纯函数 $f(z)=z+a_2z^2+\cdots$ 是满足定理条件的函数, 设 $w\in\mathbb{C}$ 但 $w\notin f(\mathbb{D})$, 则考虑函数 $1/f$, 该函数满足 *(f)*, *(c)* 中 h 的条件, 且取不到 $0,\frac{1}{w}$, 故而由 *(f)* 的结论可知

$$\left|\frac{1}{w}\right|\leq 4,$$

即有

$$|w|\geq\frac{1}{4},$$

因此有 $B(0,\frac{1}{4})\subset f(\mathbb{D})$, 定理得证.